

Cubes & dice

COMPLETE LEARNING GUIDE

What You'll Learn

Cube and dice questions test 3D spatial visualization — deducing hidden opposite faces from multiple dice views, solving cube-painting and cube-cutting problems, folding a flat net into a 3D cube, counting cubes in a stacked arrangement, and reasoning about dice rotation and mirror-identical dice. Unlike other non-verbal topics, these questions reward a small set of exact, checkable rules (the standard-die sum-of-7 rule, the common-face deduction method for non-standard dice, and the systematic corner/edge/face/internal counting formulas for cube-cutting) far more than raw spatial intuition, so memorizing and practicing the rules directly outperforms trying to visualize everything from scratch each time. This sub-module works through each rule with a complete, verified numerical example, including the classic painted-cube cutting problem worked to an exact count.

1. Standard Dice: The Sum-of-7 Opposite-Face Rule

On a STANDARD die (the kind used in board games), opposite faces always sum to 7: 1 is opposite 6, 2 is opposite 5, and 3 is opposite 4. This rule applies ONLY to standard dice — non-standard dice (with letters, symbols, or a different number arrangement) must be solved using the deduction method below instead.

2. Non-Standard Dice: The Common-Face Deduction Method

For dice with arbitrary symbols/numbers (not guaranteed to follow sum-of-7), deduce opposite faces from multiple throws using this rule: if two different throws of the same die share exactly two visible faces in common, then the two NON-common (differing) faces from those throws must be opposite each other.

- Three throws of a die show: Throw 1: faces 1, 2, 3. Throw 2: faces 1, 3, 4. Throw 3: faces 1, 4, 5. Comparing Throw 1 & 2 (common faces 1, 3): the non-common faces 2 and 4 are opposite. Comparing Throw 2 & 3 (common faces 1, 4): the non-common faces 3 and 5 are opposite. That leaves faces 1 and 6 (the only face never shown) as the remaining opposite pair.

3. Cube Painting and Cutting Problems

A classic problem: a cube is painted on all 6 outer faces, then cut into n^3 smaller identical cubes. Corner cubes (3 faces painted) = always 8 (the 8 corners of any cube). Edge cubes (2 faces painted) = $12 \times (n-2)$ (12 edges, each with $(n-2)$ non-corner small cubes along it). Face-center cubes (1 face painted) = $6 \times (n-2)^2$ (6 faces). Fully internal cubes (0 faces painted) = $(n-2)^3$.

- A cube is painted on all faces and cut into 64 smaller cubes (so $n=4$, since $4^3=64$). 3-face-painted (corners) = 8. 2-face-painted (edges) = $12 \times (4 - 2) = 24$. 1-face-painted (face centers) = $6 \times (4 - 2)^2 = 6 \times 4 = 24$. 0-face-painted (internal) = $(4 - 2)^3 = 8$. Check: $8 + 24 + 24 + 8 = 64$.

4. Folding a Net Into a Cube

For a net drawn as a straight strip of 4 connected squares (which folds into the 4 side faces of a cube, forming a loop), the 1st and 3rd squares in the strip become opposite faces, and the 2nd and 4th squares become opposite faces — because folding a 4-square strip into a closed loop wraps the 1st and 3rd (and 2nd and 4th) around to face away from each other.

5. Counting Cubes in a 3D Stacked Arrangement

For a solid $n \times n \times n$ arrangement of unit cubes with the entire outer surface painted, the same corner/edge/face/internal formulas from the cube-cutting problem apply directly, since a painted stack and a painted-then-cut cube are geometrically identical setups.

- A $3 \times 3 \times 3$ stack of 27 unit cubes has its entire outer surface painted. Corners (3 faces) = 8. Edges (2 faces) = $12 \times (3 - 2) = 12$. Face centers (1 face) = $6 \times (3 - 2)^2 = 6$. Internal (0 faces) = $(3 - 2)^3 = 1$. Check: $8 + 12 + 6 + 1 = 27$.

6. Dice Rotation Tracking

When a die rolls (tips over one of its edges) in a stated direction, track each face's new position systematically: rolling FORWARD over the front-bottom edge moves top!front, front!bottom, bottom!back, back!top (the left and right faces stay unchanged).

- A standard die shows 1 on top and 2 on the front (so, by the sum-of-7 rule, the bottom is 6 and the back is 5). After rolling forward once: new top = old back = 5; new front = old top = 1. The die now shows 5 on top and 1 on the front.

7. Identical vs Mirror-Image (Different) Dice

Two dice images showing the same three numbers around one corner represent the SAME die only if those three numbers appear in the same rotational (clockwise or anticlockwise) order in both images; if the order is reversed between the two images, they are mirror images of each other and represent genuinely DIFFERENT dice — a frequently tested trap in 'which of these dice is the same as the given one' questions.

Common Pitfalls

- Applying the sum-of-7 shortcut to a non-standard die (with letters, symbols, or unusual numbering) where it does not apply.
- In the common-face deduction method, comparing throws that share fewer or more than exactly two common faces, making the deduction invalid.

- In cube-cutting problems, forgetting the $(n^2 - 2)$ adjustment for edge and face-center cube counts, or using n instead of $(n^2 - 2)$ for internal cubes.
- Misapplying the net-folding rule to a net shape that isn't a simple 4-square strip (more complex nets need step-by-step mental folding instead).
- Tracking dice rotation by only updating the top and front faces and forgetting that the bottom and back faces also move.
- Assuming two dice images with the same three visible numbers are identical without checking whether the rotational (clockwise/anticlockwise) order actually matches.

Key Takeaways

- Standard dice: opposite faces sum to 7 (1-6, 2-5, 3-4) — this rule applies ONLY to standard numbered dice.
- For non-standard dice, use the common-face method: if two throws share exactly two faces, their non-common faces are opposite.
- Cube-cutting formulas (for an $n \times n \times n$ cut): corners = 8, edges = $12(n^2 - 2)$, face-centers = $6(n^2 - 2)^2$, internal = $(n^2 - 2)^3$.
- A 4-square net strip folds so that the 1st/3rd squares become opposite faces, and the 2nd/4th squares become opposite faces.
- Rolling a die forward cycles top! → front! → bottom! → back! → top; rolling sideways cycles the corresponding side faces instead.
- Two dice showing the same numbers are identical only if those numbers share the same rotational order — a reversed order means mirror-image (different) dice.

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